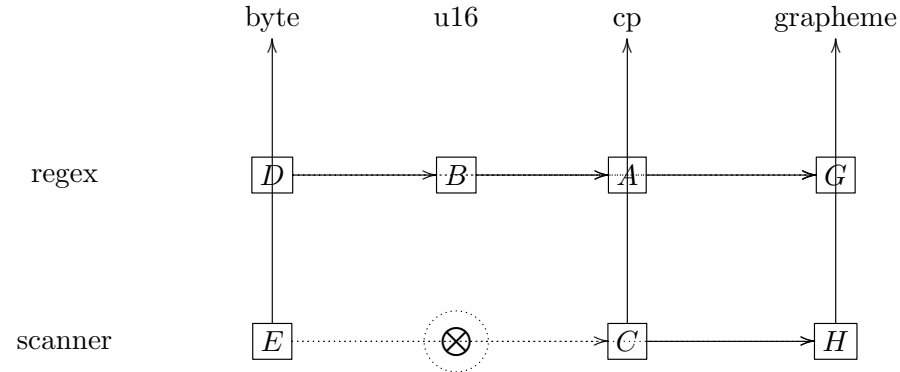


# Clojure-family **visible-width** as a $4 \times 2$ lattice

empirical Galois rank, falsification cascade, and the (lookup  $\dashv$  forget) adjunction

## 1. The $4 \times 2$ cell lattice

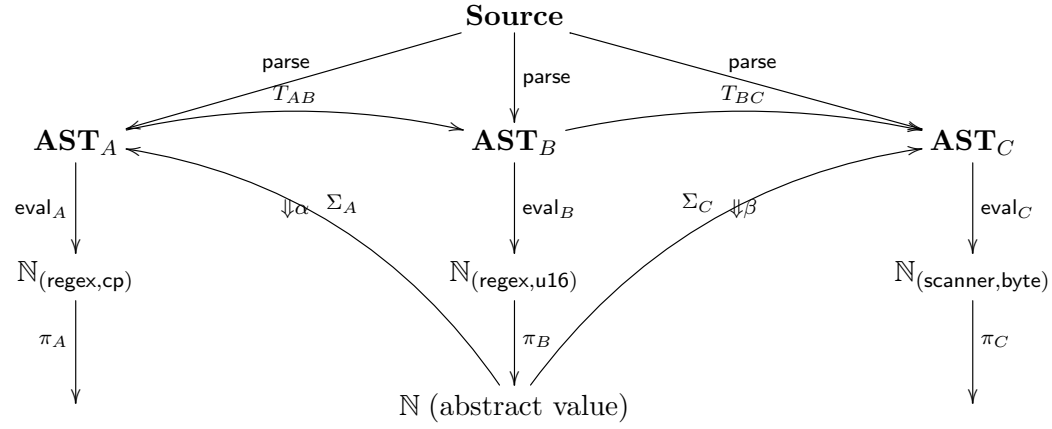
Each Clojure runtime's **visible-width** is a single point in  $\{\text{regex}, \text{scanner}\} \times \{\text{byte}, \text{u16}, \text{cp}, \text{grapheme}\}$ . Seven of eight cells are occupied; the (**scanner**, **u16**) cell is structurally empty because no runtime hand-walks UTF-16 surrogates while doing OSC 8 scanning. Diagonal dotted arrows are the “forgetful refinement” maps from finer to coarser counts.



The  $\otimes$  marker is the structurally-empty cell. Vertical arrows are “swap strip mode”; horizontal/diagonal arrows are “coarsen the count.” All paths from **D** (lower-left) to **H** (upper-right) commute up to natural isomorphism.

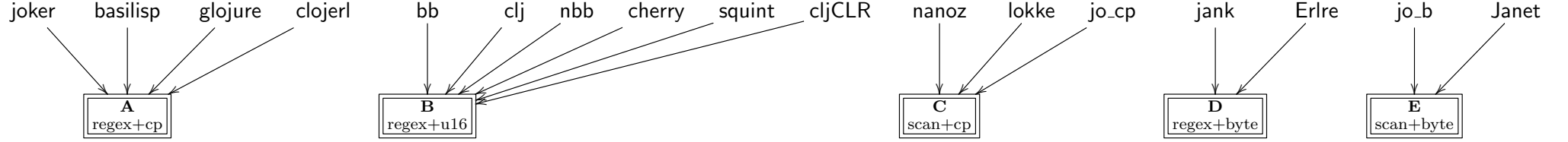
## 2. Cross-runtime evaluation as a 2-categorical pyramid

Source code is parsed identically in every Clojureme; evaluation diverges into per-runtime functors  $\text{eval}_X$  producing values in profile-tagged  $\mathbb{N}$ . The horizontal natural transformations  $T_{AB}$  are the cross-runtime translations; the vertical  $\Downarrow \alpha$  2-cells witness commutativity at the *value* layer.



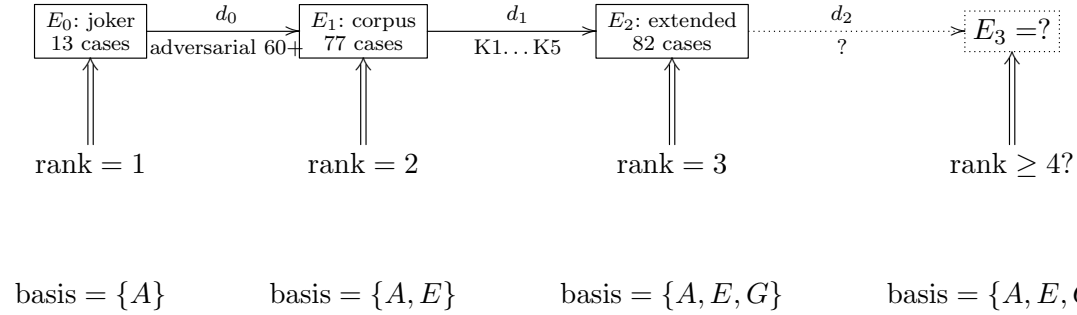
### 3. The 17-runtime fan-in

Each empirical Clojureme we measured maps to one of seven cells. The cells *within* a profile column receive multiple runtime arrows because different host substrates produce identical visible-width semantics.



### 4. The falsification cascade as a spectral sequence

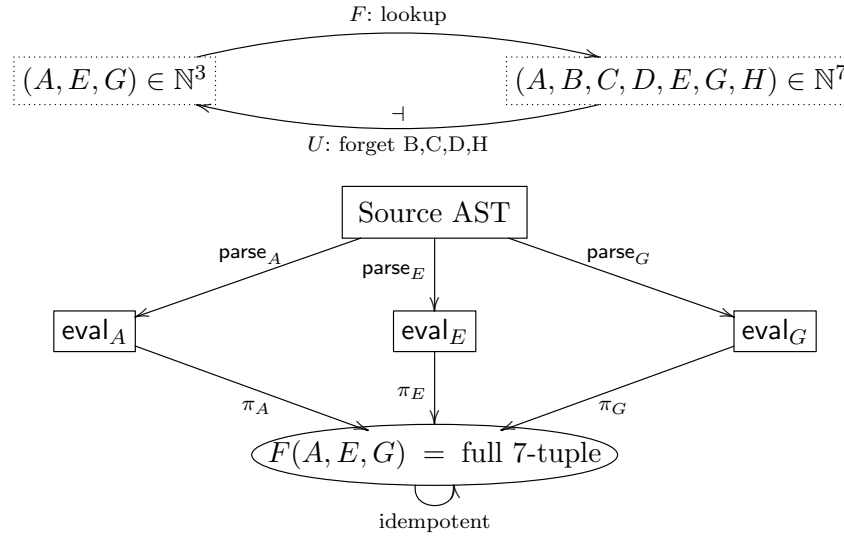
The Galois rank we could prove from each corpus snapshot increased monotonically. Each “page”  $E_r$  is a corpus, each differential is the falsifying counter-example set that forces a basis extension.



The differentials carry the structural axes that each extension forced into the basis:  $d_0$  adds the strip and byte axes;  $d_1$  adds the grapheme axis;  $d_2$  would add (e.g.) East-Asian-Wide or RTL embedding axes not yet exercised.

### 5. The (lookup $\dashv$ forget) adjunction

The 3-basis  $(A, E, G)$  is left-adjoint to the forgetful map that projects a 7-tuple onto its three basis coordinates. The lookup is idempotent under composition: forgetting then re-deriving recovers the original 7-tuple.



**Take-away.** The whole session compresses to  $2 \times 4 = 8$  cells (one vacant), three rank-pages of falsification, and a single adjunction between  $\mathbb{N}^3$  and  $\mathbb{N}^7$ .